2026 MIDDLE SCHOOL STEM CAMPS

PY-STEM

STUDENT HANDOUT PACKET

A week of team STEM missions. Predict, build, test, score, redesign, defend.

IN THIS PACKET

ACTIVITY PYS-01 · BIOMEDICAL

Magnetic Capsule Maze Cup

Steer a magnetic pill-camera through a body maze without touching it.

Win condition: Fastest clean run with low penalties and a clear magnetic-control explanation.

THE SCIENCE

Remote actuation. A magnet pulls a steel object through a barrier without contact. Doctors use the same idea to steer a swallowed capsule camera through the gut from outside the body. The field passes through the wall; only the magnetic force does the work.

Path planning. Move too fast and the capsule overshoots; move too slow and you waste time. You plan a route through the maze and control the external magnet to get a fast, clean run with the fewest wall touches.

BUILD AND RUN IT

01 **Test the field through a barrier.** Place the steel token on the maze and move a magnet under the board. Feel how the field pulls through.

02 **Plan a route.** Trace the maze path you will follow and mark tricky corners.

03 **Practice control.** Practice slow, steady moves. Sudden jerks make the token jump walls.

04 **Run for time.** Make a timed clean run. Each wall touch is a penalty.

05 **Explain the control.** Describe how distance and speed of the magnet changed your control.

Safety: You steer only the steel token (a paperclip or washer). NEVER touch or hold the tiny magnets. Swallowing two magnets, or a magnet plus a steel object, is a medical emergency. Keep all magnets away from mouths, electronics, and medical implants. A teacher counts magnets in and out. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Fastest clean run	50
Low penalties	20
Magnetic-control explanation	20
Redesign	10
Total	**100**

ACTIVITY PYS-02 · MATERIALS

Oobleck Armor Arena

Design armor with a material that acts like a liquid until impact.

Win condition: Best target protection with least material and cleanest reset.

THE SCIENCE

Shear-thickening fluids. Oobleck is cornstarch and water. Press it slowly and it flows like a liquid; hit it fast and it stiffens like a solid. Fast force jams the suspended particles together so they resist. This is shear thickening, and real impact-protection materials use the same trick.

Material efficiency. More material protects better but costs and weighs more. You optimize a sealed oobleck pad to protect a target with the least material, then test it with a controlled press, not a messy drop.

BUILD AND RUN IT

01 **Mix and feel it.** Mix cornstarch and water until it pours slowly but resists a fast poke.

02 **Seal a pad.** Double-bag a measured amount of oobleck to make a clean, sealed armor pad.

03 **Protect the target.** Place the pad over the target. Press a flat lid on top with a sharp counted "1-2-3 SLAP" (fast). For contrast, also lean slowly for 5 seconds and watch the target sink. Fast press protects; slow lean does not. Never rest a weight on the pad.

04 **Trim the material.** Reduce the oobleck while still protecting the target. Track how little you can use.

05 **Reset clean.** Show a clean reset with no leaks. Cleanup is scored.

Safety: No eating. Double-bag pads. Clean floors immediately. Goggles for splash risk. Allergy: Corn allergy note. Use sealed teacher-prepared bags if needed.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Protection result	40
Material efficiency	20
Explanation	20
Redesign	10
Cleanup	10
Total	**100**

ACTIVITY PYS-03 · MECHANICS

Cardboard Automata Arcade

Build a one-crank machine that performs a mini task.

Win condition: Most reliable motion with useful mechanism and clear explanation.

THE SCIENCE

Cams, followers, linkages. Turning a crank spins a cam. A follower rides on the cam and rises and falls as the shape passes, turning steady rotation into up-and-down or back-and-forth motion. Linkages pass that motion along. The cam shape decides the movement.

Reliability. A machine that jams is a failed machine. Smooth holes, low friction, and a steady crank make motion repeatable. You build a hand-cranked automaton that completes a task and runs reliably, then explain the mechanism.

BUILD AND RUN IT

01 **Build the box and shaft.** Make a sturdy box and pass a skewer through straw bearings so it turns freely.

02 **Cut a cam.** Cut a cam disc and glue it firmly to the shaft so it cannot slip or spin. An off-center or oval cam gives more motion.

03 **Add a follower.** Glue a straw into the top of the frame as a guide, then slide a vertical rod down through it to rest on the cam. The rod must slide freely in the straw so it rises and falls as you crank.

04 **Top it with a task.** Attach a character or tool that performs a mini task as the follower moves.

05 **Tune for reliability.** Reduce friction and wobble so it runs the same every turn.

Safety: Hot glue is staff-supervised. Scissors stay seated and closed when not in use. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Reliable motion	30
Cam or linkage complexity	20
Task success	20
Explanation	20
Aesthetics and build quality	10
Total	**100**

ACTIVITY PYS-04 · BIOMEDICAL

Stethoscope Sprint and Recovery

Build a medical listening tool and use it like a sports scientist.

Win condition: Best sound quality plus accurate, respectful data collection.

THE SCIENCE

Sound transmission. A stethoscope collects faint body sounds with a wide funnel and channels them through a tube to your ear. A sealed air path and a good funnel make quiet heartbeats audible. Materials and seal quality decide how well it works.

Heart rate and recovery. After light exercise, heart rate rises and then recovers. With consent, you measure resting and post-activity rates and watch recovery. Respectful, accurate data collection is part of the score.

BUILD AND RUN IT

01 **Build the stethoscope.** Attach a funnel to tubing and seal the joints so no air leaks.

02 **Add a diaphragm.** Stretch a balloon membrane over the funnel to pick up faint sounds, optional.

03 **Test sound quality.** Listen for a clear heartbeat. Improve the seal until it is easy to hear.

04 **Collect resting data.** With consent, count the pulse at the wrist or neck for a timed window. If the stethoscope is hard to hear, the wrist or neck count still works.

05 **Measure recovery.** After light activity, measure how heart rate returns toward rest. Sanitize shared parts.

Safety: Sanitize shared parts. No forced heart-rate measurement. Use pulse count alternative. Allergy: Latex balloon alternative: plastic wrap or nitrile membrane.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Sound quality	30
Measurement accuracy	25
Explanation	20
Redesign	15
Data clarity	10
Total	**100**

ACTIVITY PYS-05 · BIOMEDICAL

Reaction Time Combine

Train like a sports scientist and prove which strategy cuts delay.

Win condition: Best median plus most improvement using clean data.

THE SCIENCE

From signal to muscle. Reaction time is the delay between seeing a signal and moving. The eye sends a signal to the brain, the brain decides, and nerves fire the muscle. The classic ruler-drop test turns that delay into a distance you can measure.

Median and improvement. One trial is noisy, so you take many and use the median. Then you test whether a strategy, like focusing or warming up, actually improves your number. Clean data and real improvement both score.

BUILD AND RUN IT

01 **Run the ruler drop.** A partner drops a meter stick; you catch it. Read the catch distance in cm, then use the cm-to-ms strip to turn it into your reaction time.

02 **Take many trials.** Record at least ten catches. Use the median, not the best single catch.

03 **Pick a strategy.** Choose one strategy to test: focus cue, warm-up, or eyes on the release point.

04 **Retest and compare.** Run another set with the strategy. Compare medians to see if it helped.

05 **Show the data.** Present a clear before-and-after with your median and spread.

Safety: Clear lanes. No collisions. All stations can be seated. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Best median reaction time	35
Data quality	20
Strategy improvement	20
Explanation	15
Teamwork	10
Total	**100**

ACTIVITY PYS-06 · WAVES + SOUND

SONAR Slinky Showdown

Make invisible distance sensing visible with a slinky pulse.

Win condition: Most correct wave predictions and station solutions.

THE SCIENCE

Longitudinal waves. Push one end of a stretched slinky and a compression travels along it: a longitudinal wave, where the coils move back and forth along the direction of travel. Sound works the same way through air. The pulse reflects off the far end and returns.

Echo timing is ranging. If you know how fast a pulse travels and you time its round trip, you can compute distance. That is how SONAR and ultrasonic sensors measure range. You predict and interpret pulses at challenge stations.

BUILD AND RUN IT

01 **Send a pulse.** Stretch the slinky on the floor and push one end sharply to launch a compression pulse.

02 **Watch the reflection.** See the pulse travel, reflect off the held end, and come back.

03 **Predict before testing.** At each station, predict what the pulse will do, then test and check.

04 **Time several round trips.** Send one pulse, count 4 to 6 reflections, time the whole sequence, then divide the total distance by the total time. That gives a steadier speed than timing one trip, the SONAR idea.

05 **Solve the stations.** Work the challenge cards as a team, explaining each prediction.

Safety: Keep the slinky flat on the floor and never let go while it is stretched, or it can snap back and hurt someone. Do not overstretch it (about 3 m max). Keep fingers clear of coils. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Model and predictions	35
Station performance	25
Explanation	20
Team communication	20
Total	**100**

ACTIVITY PYS-07 · OPTICS + LIGHT

Pinhole Precision Challenge

Build the sharpest camera image with no lens.

Win condition: Best image clarity and concept explanation after one aperture redesign.

THE SCIENCE

Light travels in straight lines. A pinhole camera has no lens. Light from each point of a scene travels in a straight line through a tiny hole and lands on the screen, forming an upside-down image. Because the rays cross at the hole, the picture is flipped.

The aperture tradeoff. There is a BEST hole size. Too big is bright but blurry. Too tiny is dim and ALSO blurry, because light spreads out at a very small hole. Test a few sizes, find the sharpest, then redesign once.

BUILD AND RUN IT

01 **Build the viewer.** Make a tube or box with a foil aperture at one end and a wax-paper screen at the other. You will look THROUGH the wax paper toward the light.

02 **Punch a clean hole.** Pierce a clean, round hole in the foil with a sharp pin and twist it. About pinhead size to start; do not chase a hole that is as tiny as you can make it.

03 **Find the image.** Dim the room lights and aim at a bright source (a flashlight or sunny window). Look THROUGH the wax-paper screen for the upside-down image. A cut-out shape taped over the flashlight makes the flip easy to see.

04 **Tune the aperture.** Try a slightly larger or smaller hole. Balance sharpness against brightness.

05 **Explain the optics.** Describe why the image is flipped and why hole size changes the result.

Safety: No direct sun viewing. Pins handled at a controlled station. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Image clarity	35
Concept explanation	25
Build quality	20
Redesign	10
Speed	10
Total	**100**

Low-Ropes Force Map Relay

Predict balance and stability before the balance challenge proves you right.

Win condition: Most accurate predictions and clear debrief connections.

THE SCIENCE

Center of mass. Your center of mass is the average position of your weight. You stay balanced while it stays over your base of support. Lean too far and the line of gravity leaves the base, so you tip. Lowering your center of mass adds stability.

Mapping forces. At each balance challenge, you predict where balance will be hard, then test it with your body and a partner. The four challenges: (1) stand heels and back against a wall and try to lift your far foot or reach a coin on the floor ahead of you; (2) sit in a backless chair, torso straight, and try to stand without leaning forward; (3) bend, grab your toes, and try to hop forward; (4) walk a taped floor line holding a small weight at arm's length, then at your chest. Each one fails or works for a center-of-mass reason. Accurate predictions and a clear debrief score.

BUILD AND RUN IT

01 **Read the challenge.** Look at the next balance challenge card and the body position it asks for.

02 **Predict balance points.** On the card, mark whether it will work or fail, and where your center of mass moves.

03 **Test it.** Do the challenge near the wall so you can catch yourself. Notice where you actually felt unstable.

04 **Map the forces.** Sketch where your weight acted and whether your center of mass stayed over your base of support.

05 **Debrief the connection.** Explain how lowering or shifting your center of mass changed stability.

Safety: Do every challenge next to the wall with your back and heels touching it so a slip just leaves you leaning on the wall. Keep the floor clear of bags and chairs. A helper spots the chair stand. The taped line is flat on the floor: no running, no raised beam.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct predictions	35
Force-map quality	25
Debrief explanation	20
Teamwork reflection	20
Total	**100**

ACTIVITY PYS-09 · MECHANICS

Hovercraft Hockey Hackathon

Build a puck that moves by barely touching the floor.

Win condition: Best glide plus control in a gym-safe tournament.

THE SCIENCE

Air cushion cuts friction. A balloon pushes air down through a hole in a disc, lifting the disc on a thin cushion of air. With almost no contact, friction nearly vanishes, so a small push sends it gliding. Newton's laws then keep it moving until something stops it.

Glide versus control. Maximum glide is easy; control is hard. You balance lift and steerability to compete in gym-safe target hockey. Build quality and a clear explanation also score.

BUILD AND RUN IT

01 **Mount the valve.** Glue a pop-top cap over the center hole of the disc, sealing the edge.

02 **Attach the balloon.** Stretch an inflated balloon over the closed cap so air escapes only through the hole.

03 **Test the glide.** Open the cap just a little, set it down, and give a gentle push. A small opening glides longer and steers better; open wider only if it will not lift.

04 **Practice control.** Learn to start and stop on target. Too much lift means no control.

05 **Play target hockey.** Compete in gym-safe rounds. Glide and control both count.

Safety: Latex allergy note. No face-level launches. Play floor-only. A grown-up inflates the balloons; throw away any balloon that pops right away. Allergy: Latex balloon alternative: small bags or school-approved nonlatex balloons.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Glide performance	30
Target competition score	25
Explanation	20
Redesign	15
Build quality	10
Total	**100**

ACTIVITY PYS-10 · OPTICS + LIGHT

Spectra Sleuth Showdown

Can two white lights hide different color fingerprints?

Win condition: Most correct spectrum matches and strongest safe fireworks-science explanation.

THE SCIENCE

Diffraction splits light. A diffraction grating bends different colors by different amounts, spreading light into a spectrum. Two lights that look the same white can have very different spectra: a smooth rainbow from a hot filament, a broad colored band from a white LED, or separated bright lines from a neon or gas source.

Spectra as fingerprints. Each light source has a characteristic spectrum, its fingerprint. This is the same science behind firework colors, where different metal salts emit specific colors. You match mystery sources to clue cards using their spectra, no flame needed.

BUILD AND RUN IT

01 **Look through the grating.** View a white light through the grating and watch it spread into colors.

02 **Compare two sources.** Look at a white LED and a neon light. The LED shows a broad band; the neon shows separate bright lines.

03 **Sketch the spectra.** Draw what you see for each mystery source: continuous band or separate lines.

04 **Match to clue cards.** Use your sketches to match each source to its spectrum card.

05 **Connect to fireworks.** Explain how this links to the colors in a firework display, safely.

Safety: No flame tests. No lasers. Diffraction glasses are not eye protection. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct matches	40
Spectrum sketches	20
Explanation	20
Exhibit connection	20
Total	**100**

ACTIVITY PYS-11 · COMPUTING

BookBot Bin Logic Challenge

Retrieve the right book by using bin addresses, not a conveyor race.

Win condition: Most correct retrievals with few traffic conflicts and clear algorithm defense.

THE SCIENCE

Automated storage and retrieval. Temple's Charles Library BookBot stores books in bins identified by an address, not by subject on a shelf. A crane fetches the bin by its address on request. The same logic runs warehouses: store anywhere, remember the address, retrieve on demand.

Routing and search. Good addressing and smart routing minimize travel and avoid traffic jams. You simulate storage and retrieval with order cards, bin addresses, and routing rules, then defend your algorithm.

BUILD AND RUN IT

01 **Set up the bins.** Letter the rows down the side (A-D) and number the columns across the top (1-6) so an address like B3 is clear. Place items by address, not by subject.

02 **Read an order.** Draw an order card listing the items to retrieve and their addresses.

03 **Plan the route.** Order your retrievals to minimize travel and avoid collisions with other teams.

04 **Run the retrieval.** Fetch the items by address against the clock. Wrong items cost points.

05 **Defend the algorithm.** Explain the rule you used to choose the order and why it is efficient.

Safety: Keep clear of actual library operations. Tabletop version preferred after tour. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct retrievals	35
Low traffic conflicts	25
Routing or algorithm explanation	20
Efficiency	10
Teamwork	10
Total	**100**

ACTIVITY PYS-12 · DESIGN

Accessibility Ramp Rescue Lab

Design a portable ramp that works for a real user constraint.

Win condition: Best safe ramp meeting slope, load, portability, and client criteria.

THE SCIENCE

Slope, load, and universal design. A ramp trades steepness for length: a gentler slope is easier and safer to use but needs more room. Accessibility standards cap how steep a ramp can be. Universal design means building for real users and real constraints from the start, not as an afterthought.

Criteria and constraints. Your client gives criteria: a maximum slope, a load to carry, and a portability limit. You build a scale ramp for a toy wheelchair or weighted cart that meets all of them, then defend the tradeoffs.

BUILD AND RUN IT

01 **Read the client criteria.** Note the maximum slope, the required load, and the portability limit.

02 **Sketch to meet the slope.** Work out the ramp length that keeps the slope within the limit for the given height.

03 **Build the scale ramp.** Construct a sturdy ramp from foam board and supports that folds or carries easily.

04 **Test in two steps.** First, roll the cart down the ramp to check it runs smooth and straight (do not pile weights on the cart). Then bridge the ramp between two desks and hang the weights at the middle to see if the deck sags. If it sags, add dowel or craft-stick ribs underneath and retest.

05 **Defend the tradeoffs.** Explain how your design meets slope, load, and portability together.

Safety: No standing on ramps. Loads are small tabletop weights only. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Load held	30
Slope meets target	30
Portability	20
Client criteria explanation	20
Total	**100**

BACKUP ACTIVITIES

If weather or materials change

ACTIVITY PYB-01 · MECHANICS

Pencil Pressure Safety Lab

Spread a force before it dents the target.

Win condition: Largest area protected with clear data.

THE SCIENCE

Pressure vs force. Pressure is force divided by area. The same push on a sharp point dents a surface, but spread over a wide area it does not. That is why a sharp pencil pierces and a flat board does not: same force, different area.

Spreading stress. Safety design often means spreading a load so no single spot takes too much. You design a spreader that protects foam from a standard load and show the protected area with data.

BUILD AND RUN IT

01 **Test the bare point.** Lay graph paper over the foam. Press the same load straight down through a pencil point. Record whether the point pierces or tears the paper (yes/no).

02 **Design a spreader.** Build a cardboard spreader that distributes the same load over more area.

03 **Apply the standard load.** Press the same weight through your spreader onto fresh foam.

04 **Measure protected area.** Trace the spreader footprint on the graph paper and count the squares. No pierce over more squares means lower pressure.

05 **Redesign for more spread.** Improve the spreader to protect a larger area and re-test.

Safety: No skin tests. Use foam only. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Pressure spread	40
Explanation	25
Data quality	20
Redesign	15
Total	**100**

ACTIVITY PYB-02 · BIOMEDICAL

Domino Neuron Relay

Make a nerve signal survive a gap.

Win condition: Fastest reliable signal with best explanation.

THE SCIENCE

Signals travel in a chain. A line of falling dominoes is a model of a nerve signal: each one knocks the next, so the signal travels without any single domino moving far. A real neuron passes a signal down its length the same way, one step triggering the next.

Gaps and insulation. If the gap between dominoes is too large, the signal dies. Neurons face the same problem at synapses. A myelin sheath speeds signals along; you model gaps and insulation to keep the relay fast and reliable.

BUILD AND RUN IT

01 **Build a baseline track.** Tape paper down first so tiles cannot slide back. Set dominoes evenly with each gap smaller than a tile is tall (about half a tile), then time one full run.

02 **Introduce a gap.** Open a larger gap and see whether the signal still crosses.

03 **Bridge the gap.** Add a bridge piece, like a synapse helper, so the signal survives.

04 **Speed it up like myelin.** Tape a few tiles into rigid blocks that fall as one unit, so the signal jumps block to block and reaches the end faster than the evenly spaced line.

05 **Explain the analogy.** Connect your design choices to neurons, synapses, and myelin.

Safety: Low risk. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Transmission success	35
Gap challenge	25
Explanation	20
Redesign	20
Total	**100**

ACTIVITY PYB-03 · MECHANICS

Pulley Rescue Relay

Lift the load with smarter force, not stronger arms.

Win condition: Best lift with correct setup and explanation.

THE SCIENCE

Mechanical advantage. A single fixed pulley just changes the direction of your pull. Add movable pulleys and you trade distance for force: you pull more rope but with less effort. More supporting rope segments means more mechanical advantage.

Force and direction. Pulleys let you pull down to lift up, and let a small force raise a big load. You build pulley systems, compare the effort needed, and explain why the setup helps.

BUILD AND RUN IT

01 **Lift with no pulley.** Measure the effort to lift the load by hand as a baseline.

02 **Add a fixed pulley.** Reroute over one fixed pulley. Note that direction changes but effort does not.

03 **Add a movable pulley.** Build a system with a movable pulley and feel the effort drop.

04 **Measure the advantage.** Use a spring scale to compare effort across setups. Keep the same load heavy enough that the scale reads clearly, not near zero.

05 **Explain the tradeoff.** Show how more rope segments cut the force you need.

Safety: Keep feet clear of loads. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Setup accuracy	25
Explanation	20
Teamwork	20
Total	**100**

ACTIVITY PYB-04 · COMPUTING

Barcode Checksum Rescue

Catch the fake barcode before the warehouse ships the wrong part.

Win condition: Most bad codes detected with few false alarms.

THE SCIENCE

Check digits catch errors. A barcode includes an extra check digit computed from the others by a fixed rule. If a digit is mistyped or smudged, the rule no longer matches and the scanner rejects the code. This is how real product codes catch errors automatically.

Detect without false alarms. A good detector flags the truly bad codes while passing the good ones. You race to find corrupted codes, keep false alarms low, and explain the check-digit rule you used.

BUILD AND RUN IT

01 **Learn the rule.** Study the check-digit rule: how the last digit is computed from the rest.

02 **Verify a good code.** Apply the rule to a valid code and confirm it checks out.

03 **Hunt corrupted codes.** Scan a deck of cards and flag the ones that fail the rule.

04 **Avoid false alarms.** Double-check before flagging; wrongly rejecting a good code costs points.

05 **Explain the rule.** Describe how the check digit catches a single mistyped digit.

Safety: Normal classroom management. Allergy: None.

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct detections	45
Low false alarms	20
Rule explanation	20
Teamwork	15
Total	**100**